

Finance theory and modeling

Lecture 7

Additional options topics

(Chapter 12, Luenberger)

Dr. L. Dam

Dr. A. Tsvetkov

Assignment 2

- We will try to grade your assignment as soon as possible.
- We cannot guarantee the grade before the exam.
- The date for perusal will be posted on Nestor.

Introduction

- Using binomial trees is straightforward and simple, and works reasonably well.
- So it is by itself sufficient to solve most option problems.
- However, a continuous time-version of the theory leads to new insights, allows for more complex derivatives, provides alternative computational methods, and gives a more complete theory of investment: Black-Scholes
- Historically Black-Scholes came actually before the Binomial tree; the tree theory is a result of simplification.

- Today is going to be slightly more complicated compared to previous weeks.
- It is mainly meant to open the door for Quantitative Finance, which is taught more extensively in the EOR Master.
- Don't feel overwhelmed. Try to follow the logic of the arguments, go slowly through the algebra yourself.
- If you get the logic of the tree, all this really boils down to is mathematical rigor.

THE MOST INFLUENTIAL FINANCE THEORY EVER

Assumptions of BS

- The price of a stock is governed by GBM:

$$dS = \mu S dt + \sigma S dz$$

- The value of a risk-free bond carrying interest rate r is governed by:

$$dB = rB dt$$

- There is a derivative to S , and its price is a function of S and t , so $f(S, t)$. (that this is true is not obvious per se) We want to know this function f .

The Black-Scholes equation

Black-Scholes Equation: A derivative of S facing interest rate r has a price $f(S, t)$ which satisfies the following equation:

$$\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} rS + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 = rf$$

(Proof will come later...)

Verifying BS for simple cases

Consider as a derivative the *underlying stock itself*; it is silly but it is a derivative of S . In this case $f(S, t) = S$ should satisfy the equation:

$$\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} rS + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 = rf$$
$$0 + 1 \cdot rS + \frac{1}{2} \cdot 0 \cdot \sigma^2 S^2 = rS$$

Check!

Verifying BS for simple cases

Consider as a derivative the bond; it also a derivative of S . In this case $f(S, t) = e^{rt}$ should satisfy the equation:

$$\begin{aligned}\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} rS + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 &= rf \\ r e^{rt} + 0 \cdot rS + \frac{1}{2} \cdot 0 \cdot \sigma^2 S^2 &= r e^{rt}\end{aligned}$$

Check!

The Black-Scholes equation

- There are *uncountable many* solutions to the partial differential equation:

$$\frac{\partial f}{\partial t} + \frac{\partial F}{\partial S} rS + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 = rf$$

- To pin down the unique price associated with the derivative, we need to specify the boundary condition.
- For a stock this is $f(S, T) = S(T)$ and a bond $f(S, T) = e^{rt}$.

Boundary conditions

- For options, we also know the boundary conditions:
 - For a call option, $f(S, t) = C(S, t)$, these are:
 - 1 $C(0, t) = 0$
 - 2 $C(S, T) = \max(S - K, 0)$
 - For a put option, $f(S, t) = P(S, t)$, these are:
 - 1 $P(\infty, t) = 0$
 - 2 $P(S, T) = \max(K - S, 0)$
- The boundary conditions pin down unique solutions for f .

Recap Black Scholes equation

- ① Assumption that stock is GBM, r is constant.
- ② *Any* derivative of the stock should satisfy the BSE, many solutions exist (since many derivatives are possible).
- ③ The boundary conditions of a particular asset determine the unique solution for f .

Proof of BS equation

The proof is by replication, in the spirit of the simple 1-period binomial tree.

- ① Since the derivative price f is a function of S , and t , we can use Ito's lemma to characterize df .
- ② We then make a portfolio G consisting of S and B and choose the weights such that the portfolio dynamics, dG , correspond to derivative dynamics, df .

Proof of BS equation

- ① According to Ito's lemma:

$$df = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} S\mu + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 \right) dt + \frac{\partial f}{\partial S} \sigma S dz.$$

- ② Now we make a (self-financing) portfolio, investing x_t in S and y_t in B at each time t . (so we change the weights over time)

So $G = xS + yB$, and if it is self-financing:

$$dG = x dS + y dB \text{ (because } S dx + B dy = 0; \text{ Why?)}$$

$$dG = x (\mu S dt + \sigma S dz) + y rB dt$$

Proof of BS equation

Rewriting gives

$$dG = (x \mu S + y r B) dt + x \sigma S dz$$

We had for the derivative:

$$df = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} S \mu + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 \right) dt + \frac{\partial f}{\partial S} \sigma S dz.$$

We want to match the dz -term, so:

$$x = \frac{\partial f}{\partial S}$$

(=delta hedging; will make sense later)

Proof of BS equation

We thus have $x = \frac{\partial f}{\partial S}$.

But we also need

$$f = G$$

$$G = xS + yB$$

So we can solve for y :

$$y = \frac{1}{B} \left[f - S \frac{\partial f}{\partial S} \right]$$

We know x and y , so plug it into:

$$dG = (x\mu S + y rB) dt + x\sigma S dz$$

Proof of BS equation

This gives

$$dG = \left(\frac{\partial f}{\partial S} \mu S + \frac{1}{B} \left[f - S \frac{\partial f}{\partial S} \right] rB \right) dt + \frac{\partial f}{\partial S} \sigma S dz$$

Simplifying gives:

$$dG = \left(\frac{\partial f}{\partial S} \mu S + r f - r S \frac{\partial f}{\partial S} \right) dt + \frac{\partial f}{\partial S} \sigma S dz$$

The derivative again:

$$df = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} S \mu + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 \right) dt + \frac{\partial f}{\partial S} \sigma S dz.$$

The drift (dt -term, in boldface) of both equations should also match...

Proof of BS equation

$$\frac{\partial f}{\partial S} \mu S + r f - r S \frac{\partial f}{\partial S} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} S \mu + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 \Rightarrow$$
$$r f - r S \frac{\partial f}{\partial S} = \frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2$$

Rewriting this equation gives the result:

$$r f = \frac{\partial f}{\partial t} + r S \frac{\partial f}{\partial S} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2$$

Note that μ , which is so hard to measure empirically, is absent in this equation. This is part of the reason why we can value derivatives so well...

Black Scholes formula

- We cannot always solve the BSE.
- But for European options there is an explicit solution.
- Let $N(x)$ be the standard **cumulative normal probability distribution**.
- **Black Scholes Call Option Formula:**

$$C(S, t) = SN(d_1) - Ke^{-r(T-t)} N(d_2)$$

$$d_1 = \left[\ln\left(\frac{S}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)(T - t) \right] / (\sigma\sqrt{T - t})$$

$$d_2 = \left[\ln\left(\frac{S}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)(T - t) \right] / (\sigma\sqrt{T - t}) = d_1 - \sigma\sqrt{T - t}$$

- Again, note there is no μ in the formula.

Black Scholes formula

$$C(S, t) = SN(d_1) - Ke^{-r(T-t)} N(d_2)$$

$$d_1 = \left[\ln\left(\frac{S}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)(T - t) \right] / (\sigma\sqrt{T - t})$$

$$d_2 = \left[\ln\left(\frac{S}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)(T - t) \right] / (\sigma\sqrt{T - t}) = d_1 - \sigma\sqrt{T - t}$$

- d_1 and d_2 depend on $\ln\left(\frac{S}{K}\right)$; $\ln\left(\frac{S}{K}\right) > 0$ if $\frac{S}{K} > 1$.
- We can see that when $t = T$,

$$d_1 = d_2 = +\infty \text{ if } S > K, \text{ and } d_1 = d_2 = -\infty \text{ if } S < K$$

- And so $N(d_1) = 1$, and $N(d_2) = 1$ if $S > K$,
- and $N(d_1) = 0$, and $N(d_2) = 0$ if $S < K$ (N is a CDF).
- Hence

$$C(S, T) = S - K \text{ if } S > K, \quad C(S, T) = 0 \text{ if } S < K.$$

Example

Suppose $K = \$60$, $S_t = 62$, $r = 10\%$, $\sigma = 20\%$, option expires in 5 months and (r and σ are per year).

$$d_1 = \frac{\ln(\frac{62}{60}) + 0.12 \cdot \frac{5}{12}}{0.2 \sqrt{\frac{5}{12}}} = 0.64$$

$$d_2 = d_1 - 0.2 \sqrt{\frac{5}{12}} = 0.51$$

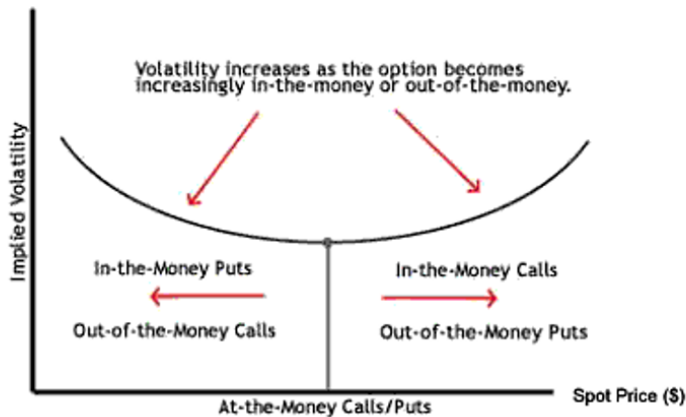
$$N(d_1) = 0.74, \quad N(d_2) = 0.70$$

$$\begin{aligned} \text{So } C(S, t) &= 62 \cdot 0.74 - 60 \cdot e^{-0.1(5/12)} \cdot 0.70 = \\ &= \$5.8 (> S - K = \$2) \end{aligned}$$

Implied volatility and smiles

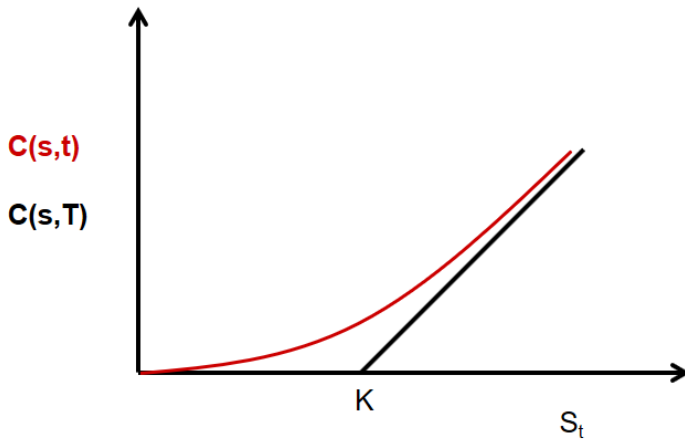
- Sometimes the formula is used the other way around.
- Given the price of an option, what is the volatility that makes the BS formula hold?
- This is called **implied volatility**.
- Implied volatility has also demonstrated the weakness in the theory.
- It should be the same, irrespective of the strike price if the option is on the same stock, same expiration date; (or equivalently, it should not depend on *how far in the money* the option is)

(implied) volatility smile



Probable reason: returns are not normal, fat tails...

Visualisation of option price



Risk neutral valuation

- We can use risk-neutral probabilities in this framework as well (In the tree we also moved from replication to risk-neutral probabilities)
- We normally have:

$$dS = \mu S dt + \sigma dz, \text{ it followed } E[S(t)] = S(0)e^{\mu t}$$

- Let us define $dw = \frac{(\mu-r)}{\sigma} dt + dz$

$$z(t) \sim \mathcal{N}(0, t), \text{ but } w(t) \sim \mathcal{N}\left(t \cdot \frac{\mu-r}{\sigma}, t\right)$$

- We simply "use dw as we did with dz " and we "pretend" outcomes are s.t. these new probabilities.

We now have:

$$dS = rS dt + \sigma S dw, \text{ with } \hat{E}[S(t)] = S(0)e^{rt}$$

Risk neutral valuation

$$\hat{E}[S(t)] = S(0)e^{rt} \text{ means: } S(0) = e^{-rt}\hat{E}[S(t)]$$

So we simply use the risk-free rate to discount the expected value (using the "new" probabilities).

It turns out that we can then value options the same way:

$$C = e^{-rT}\hat{E}[\max(S(T) - K, 0)]$$

Now use that $S(T)$ is lognormal, with

$$E\left[\ln\left(\frac{S(T)}{S(0)}\right)\right] = rT - \frac{1}{2}\sigma^2 T \text{ and } \text{Var}\left[\ln\left(\frac{S(T)}{S(0)}\right)\right] = \sigma^2 T$$

Risk neutral valuation

- So we need to evaluate $\hat{E}[\max(S(T) - K, 0)]$ and we know the distribution of $S(T)$.
- Basically it is calculating an expectation for $\max[X - K, 0]$, where we know the pdf $f(X)$ of X (lognormal in this case, so support $0, \infty$).

$$\hat{E}[\max(X - K, 0)] = \int_0^K 0f(X)dX + \int_K^\infty f(X)XdX - K \int_K^\infty f(X)dX$$

- Discounting this expectation gives eq (13.21) on page 357.
- It is the BS formula, but in integral form.

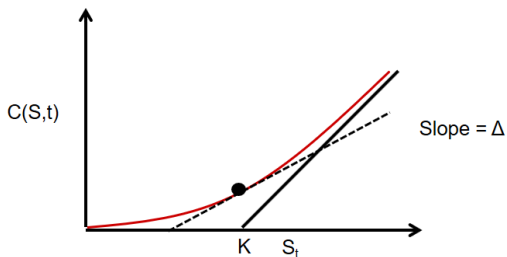
- This approach works more generally,
- so even if we have a derivative that pays X^2 at time T .
- Just calculate $\hat{E}[X^2]$ and discount back ($e^{-r(T-t)}$).
- In this case it will be proportional to the variance of X minus the price squared (Why?)
- For a bond we can check easily whether the equation makes sense:

$$\begin{aligned} B(t) &= e^{-r(T-t)} \hat{E}[B(T)] = e^{-r(T-t)} \hat{E}[B(t)e^{r(T-t)}] = \\ &= e^{-r(T-t)} B(t) e^{r(T-t)} = B(t) \end{aligned}$$

- Again, note that choosing $\hat{E}$ is nothing else than changing the probabilities (as in the binomial tree).
- So that we can discount using the risk-free rate.
- ($e^{-r(T-t)}$ in continuous time vs $\frac{1}{(1+r)}$ in the static case)
- The real nice "trick" is that to change the probabilities in "the right way" is that we only have to change the drift term, and not the diffusion term, in going from dz to dw .

Delta, gamma, theta

- The derivative of the value of security with respect to the underlying is called *Delta* (Δ).
- So $\Delta = \frac{\partial f(S,t)}{\partial S} \approx \frac{\Delta f}{\Delta S}$



Delta, gamma, theta

- Δ for a call option is $N(d_1)$.
- You can use delta to hedge risk.
- If you think options are overpriced, you may want to sell one.
- However you may not want to run any risk: of course if the stock increases in value you may lose a lot of money.
(Why? Answer: you are forced to deliver a stock at a lower price K , but to obtain it you may have to pay the higher price S_T for it).
- So you want to buy some stocks now to offset this risk... (Stock price goes up? you lose on the option but gain on the stock itself...)
- *How much* to invest in stocks? Answer: Δ

- If we have a portfolio of derivatives, we can also calculate the **portfolio delta**, which is the sum of the deltas of each component in the portfolio.
- The investor on the previous page had an investment equal to $-C(S, t) + \Delta S$.
- (sold a call and bought Δ shares).
- The derivative of S wrt $S = 1$, so delta of $S = 1$.
- Hence delta of the portfolio is $-\Delta + \Delta = 0$.
- This portfolio is **delta neutral**.

- Since the slope changes continuously as the stock price changes, we need to adjust the amount we invest in stocks all the time:
- We say we need to **rebalance** to keep the portfolio delta neutral.
- This process is a **dynamic hedging strategy**.
- We can do even better by approximating the curvature and time derivative of the figure...

- Similarly we can define:

① **Gamma** = $\frac{\partial^2 f(S,t)}{\partial S^2}$ (Γ , Curvature)

② **Theta** = $\frac{\partial f(S,t)}{\partial t}$ (Θ , time change in value)

- Using these we can approximate changes in f by:

$$\partial f \approx \Delta \partial S + \frac{1}{2} \Gamma (\partial S)^2 + \Theta \partial t$$

- If you paid attention, you may have noticed that this is an approximation of the BSE.
- You can use this to dynamically hedge your portfolio really well...

Replication, synthetic options and portfolio insurance

- The basic strategy in all **arbitrage pricing** is to **replicate** the derivative security.
- You do this by constructing a **self-financing** portfolio consisting of the underlying asset and the risk-free asset.
- We call that portfolio a **synthetic** derivative.
- It usually requires continuously rebalancing of the portfolio weights (amount invested in the underlying asset and risk-free asset change).

Replication, synthetic options and portfolio insurance

- Given no **arbitrage** the price of the synthetic security and real security should be the same.
- If not you can try to exploit this arbitrage opportunity.
- You can also use a synthetic security in combination with your portfolio to hedge against risk.
- Let's say you want to hedge against a crisis, you can use put options.
- But if these are not available, you can make a synthetic one; this is labeled **portfolio insurance**.

Replication, synthetic options and portfolio insurance

- So no arbitrage and replicating portfolios lie at the heart of most financial engineering techniques.
- It works quite well, since the important information is already in the underlying assets.
- It is **relative pricing**: a 2 liter cola bottle equals two 1 liter cola bottles, assuming the bottles away (=transaction costs etc.)
- **Absolute pricing** is harder (e.g. CAPM, APT). Here, you want to know what actually constitutes the liter price of cola in the first place.

Replication, synthetic options and portfolio insurance

- One exhibit of why relative pricing works well is that the expected risk premium $E[r_i - r_f] \mu_i - r_f$ plays no role in derivative pricing.
- Besides the fact that empirically this is hard to measure (mean blur)...
- ...the other advantage is that $\mu_i - r_f$, in a deeper sense, depends on risk-aversion and expectations of investors.
- We cannot look into peoples' heads, so figuring out the price/expected return of a stock is difficult.
- However, we do observe what they actually paid for a stock.
- So *given* that price, we can price other stuff related to that stock (such as derivatives).
- Discussions about rationality are meaningless. Arbitrage Pricing can never answer these.

Replication, synthetic options and portfolio insurance

- The availability of data, the competitiveness in financial markets - that leads to no arbitrage – and the techniques available make it almost an exact science; observe the accuracy of derivative pricing.
- For this reason, financial economics has attracted many physicists and engineers (*“Hey, the equations are the same!”*)
- However, I want you as econometricians to stay ahead of the game, go beyond financial engineering:
- Prices are not “simply smoke moving through the air”, but a result of supply and demand.

Monte Carlo simulation

- An alternative to the binomial tree approach.
- Instead of modeling all possible outcomes, you generate a lot of sample paths for S using dS .
- It is particularly useful to do sample paths for the risk-free world dynamics.
- To know what is $\hat{E}[f(S(T))]$, you simply evaluate $f(S(T))$ for all realized paths, and take the average.
- So even if we cannot get an explicit mathematic expression, we can approximate it numerically.
- This is how people also price derivatives in practice.
- You can make dynamics very complicated/realistic

- Backward induction technique on a grid.
- *Read this section once*
- My personal take on this technique is that it is often more appropriate for stochastic optimal control problems, or exotic, path-dependent derivatives...
- Trees and Monte Carlo simulations are already two good methods, which probably make more sense.
- (Plus I think Luenberger is not very clear in this subsection).

Trinomial lattices/trees

- Similar to binomial trees, uses u , 1 , d .
- The setup is a little bit more sophisticated, and thus more accurate given the same number of steps in a binomial tree.
- The recombining of the nodes is such that $ud = 11$, $du = 11$, $1d = d1$, $u1 = 1u$ (See figure 13.5).
- E.g. in 4 steps we have 9 outcomes instead of 4, which accounts for the increased precision.
- Turns out to be useful for the risk-neutral approach.

- The book lists a lot of different options,
- E.g. compound options, which are options on options
- Or knock-out options, which do not have a specific maturity date, but terminate when the price of the asset reaches a certain point (similar to Turbos or Sprinters).
- We can value all, some are easier to value than others; not always analytically, but at least numerically...
- The list does not stop where Luenberger stops...

SKIP MOST OF IT

- The relevant thing we want you to know is that dividend payments matter.
- Why? Simply because prices drop after dividend payments.
- It may be worthwhile to exercise an American Call prematurely in this case to collect the dividends; plus it changes the price path.

- The written exam will be a reflection of the tutorial exercises.
- Exam questions 1.1, 2.1, 3.1, 4.1, 5.1 comprise a past exam.
- Exam questions 1.2, 2.2, 3.2, 4.2, 5.2 also comprise a past exam.
- The structure of the exam will be the same.
- So the focus will be on calculations.
- However, we may ask a few qualitative "insight" questions...

- As said, the course QF in the Econometrics master continues the arbitrage pricing approach...
- If you are considering the ReMa: I also teach “Asset Pricing”. (Book: Cochrane)
- *Please fill out the course evaluations.*